

HADI AHMAD

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EDUCATION

UNIVERSITY OF MINNESOTA – TWIN CITIES

Bachelor of Science in Computer Science and Electrical Engineering

University Honors Program | Dean's List | 3.75 Technical GPA

Relevant Coursework: Parallel Programming, Microcontrollers, Computer Architecture, Operating Systems, Digital Design, Data Modeling, Machine Learning, Signals Circuits & Electronics, Database Systems

Awards: Presidential Scholarship, Iron Range Scholarship, Dakota Electric Association Scholarship

Minneapolis, MN

Expected May 2028

PROJECTS

PLANUMN | *React, Next.js, TypeScript, PostgreSQL, SQLite* | planu.mn

April 2025 – August 2025

Graduation Planner Web Application

- Launched a drag-and-drop, mobile-adaptive graduation planning tool used by **80+** UMN students pre-release.
- Deployed a transcript import feature that parses unofficial UMN PDFs to autofill completed courses.
- Integrated a **PostgreSQL** schema via **Supabase**, with user authentication, RLS, and server-side permission checks.
- Built responsive UI components using **Next.js** and **TypeScript**; server-side **SQLite** for fast course data fetching.

ANALOG AUDIO SYSTEM | *Active Filters, FFT Analysis, Analog Circuits* | **Course Project**

March 2026 – May 2026

Variable Gain, Bass & Treble Audio Amplifier

- Designed and breadboarded a three-stage LM741-based analog audio amplifier circuit — variable-gain, bass tone-control, and treble tone-control — enabling continuous gain and tone shaping of any 3.5mm audio source.
- Swept a potentiometer-controlled feedback network to characterize the VGA's full gain range, then used **frequency response analysis** to confirm the tradeoffs caused by the 1 MHz gain-bandwidth product of the LM741.
- Modeled a capacitor-tuned bass shelving filter, deriving boost/cut behaviour from feedback theory, then validated the design by swapping timing capacitors to observe the relationship between capacitance and corner frequency.

KINTSUGI | *Python, Web-scraping, Google Maps API, Next.js* | [placominihack](https://placominihack.com)

22 hours in Feb 2026

Global Conflict Awareness Heatmap (MNHack '26 Submission)

- Visualized global conflict on an interactive heat map using **web-scraped** Twitter data and user-submitted reports.
- Built **Python Natural Language Processing** pipeline leveraging a **SentenceTransformer** and Pandas to process tweet datasets, classify event severity through LLM prompting, and extract geolocation from unstructured text.
- Pitched at MNHack 2026 to address centralized media control by proposing an equal-visibility model that surfaces underreported crises independent of behavioural algorithms and commercial interests.

GENE EXPRESSION ANALYSIS | *Tucker Decomposition, MATLAB* | **Course Project**

April 2025 – May 2025

Colon Cancer Gene Expression Project

- Built a tensor decomposition pipeline with **MATLAB**'s Deep Learning plugin to analyze spatial transcriptomics data
- Applied **GraphTucker** to extract low-rank spatial features from 3D gene-spot tensors, for unsupervised clustering.
- Derived biologically relevant insights; Identified clusters associated with immune response, chemoresistance, and tumor growth using PPI networks and BIOGRID datasets. Presented results with biological interpretations.

PROFESSIONAL EXPERIENCE

RETAIL / FOOD SERVICE EXPERIENCE

Lakeville, MN

McDonalds • Blaze Pizza • Planet Fitness • Target • Walmart

Sep 2022 – Present

- Consistently completed time-based and recurring operational tasks (hourly facility checks, daily cleaning cycles)
- Handled high-pressure, high-volume situations with efficiency (long lines, equipment failures, customer altercations)

TECHNICAL SKILLS

LANGUAGES: Python, Java, JavaScript/TypeScript, C/C++, Verilog, Swift, SQL, HTML/CSS, MATLAB

TECHNOLOGIES: Git/GitHub, React/React Native, Node.js, AWS, Docker, LangChain, Playwright

DESIGN: KiCAD, Vivado, Simulink, LTSpice **INSTRUMENTATION:** Spectrum Analyzer, Oscilloscope, DMM